COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating System Administration	**SECTION:**	CPE31S2
DATE PERFORMED:	8-21-26	**DATE SUBMITTED:**	8-21-26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

Task 1:

1.

```
 Failed to lock directory /var/lib/apt/lists/: E:Could not open lock file /var/l
ib/apt/lists/lock - open (13: Permission denied)
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'update_cache': 'true'}, 'timeout': 0, 'async_val': 0
, 'poll': 15}

ubuntu1 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
 /var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
(13: Permission denied)"
}
ubuntu2 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
 /var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
```

What is the result of the command? Is it successful?

- It failed and no it was not successful.

```
masongsong@Masongsong:~/HOA4.1$ ansible all -m apt -a update_cache=true --become
 --ask-become-pass
BECOME password:
ubuntu2 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787287385,
    "cache_updated": true,
    "changed": true
}
ubuntu1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787287386,
    "cache_updated": true,
    "changed": true
}
masongsong@Masongsong:~/HOA4.1$
```

2.

A.

```
masongsong@Masongsong:~$ which vim
/usr/bin/vim
masongsong@Masongsong:~$ apt search vim-nox
Sorting... Done
Full Text Search... Done
vim-nox/noble-security,now 2:9.1.0016-1ubuntu7.19 amd64 [installed]
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/noble-security,now 2:9.1.0016-1ubuntu7.19 amd64 [installed,automatic]
  Vi IMproved - enhanced vi editor - compact version
```

- Both commands were successful.

B.

```
Start-Date: 2026-08-21  03:40:33
Commandline: aptdaemon role='role-commit-packages' sender=':1.119'
Install: linux-hwe-7.0-tools-7.0.0-30:amd64 (7.0.0-30.30~24.04.1, automatic), linux-mo>
Upgrade: linux-headers-generic-hwe-24.04:amd64 (7.0.0-28.28~24.04.1, 7.0.0-30.30~24.04>
End-Date: 2026-08-21  03:45:23

Start-Date: 2026-08-21  04:52:52
Commandline: apt install vim-nox
Requested-By: masongsong (1000)
Install: ruby-sdbm:amd64 (1.0.0-5build4, automatic), liblua5.1-0:amd64 (5.1.5-9build2,>
End-Date: 2026-08-21  04:53:09
```

- I see all of the commands that I used to install different applications and packages.

3.

A.

```
masongsong@Masongsong:~/HOA4.1$ ansible all -m apt -a name=snapd --become --ask-become-
pass
BECOME password:
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787287385,
    "cache_updated": false,
    "changed": false
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787287386,
    "cache_updated": false,
    "changed": false
}
masongsong@Masongsong:~/HOA4.1$
```

Can you describe the result of this command? Is it a success? Did it change anything in the remote servers?

- Cache_updated and changed were false when I last inputted ansible all -m ping. The command was a success and it outputs that both cache_updated and changed are not false.

B.

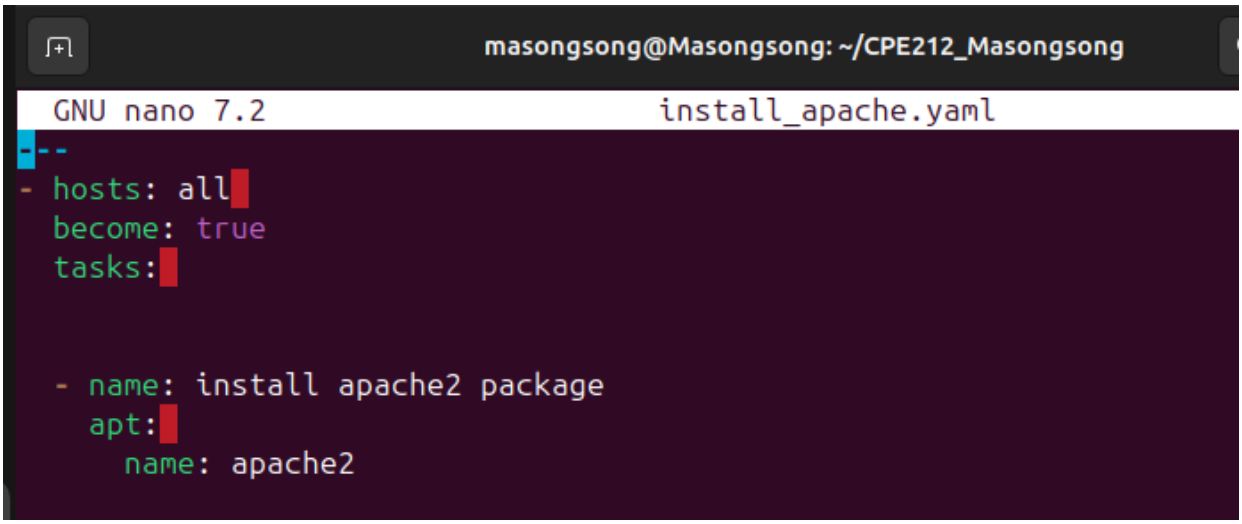
```
masongsong@Masongsong:~/HOA4.1$ ansible all -m apt -a "name=snapd state=latest" --becom
e --ask-become-pass
BECOME password:
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787287385,
    "cache_updated": false,
    "changed": false
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787287386,
    "cache_updated": false,
    "changed": false
}
masongsong@Masongsong:~/HOA4.1$
```

Describe the output of this command. Notice how we added the command state=latest and placed them in double quotations

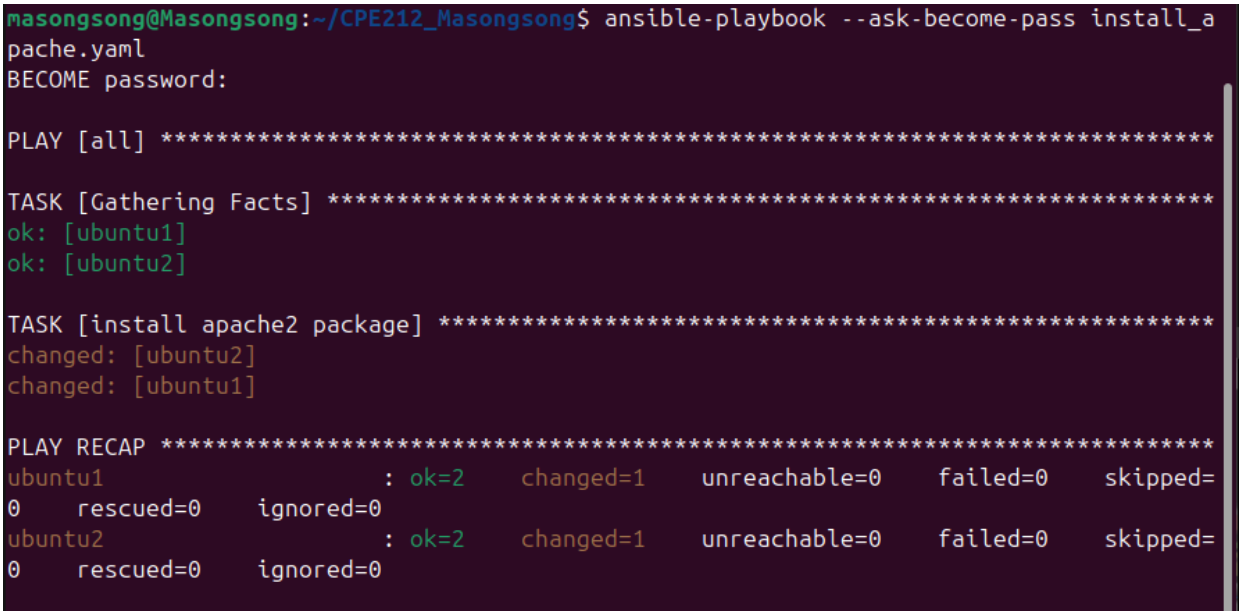
- It just lets us install the latest version of snapd package.

Task 2:

1.

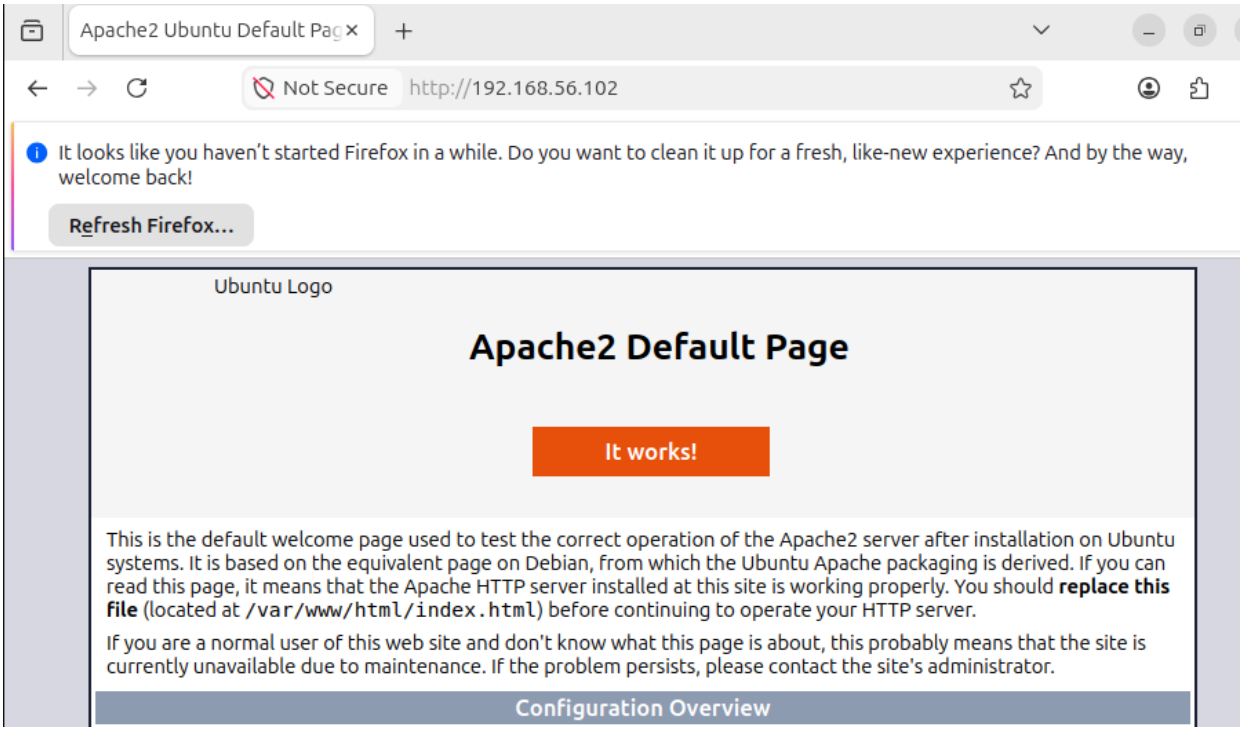


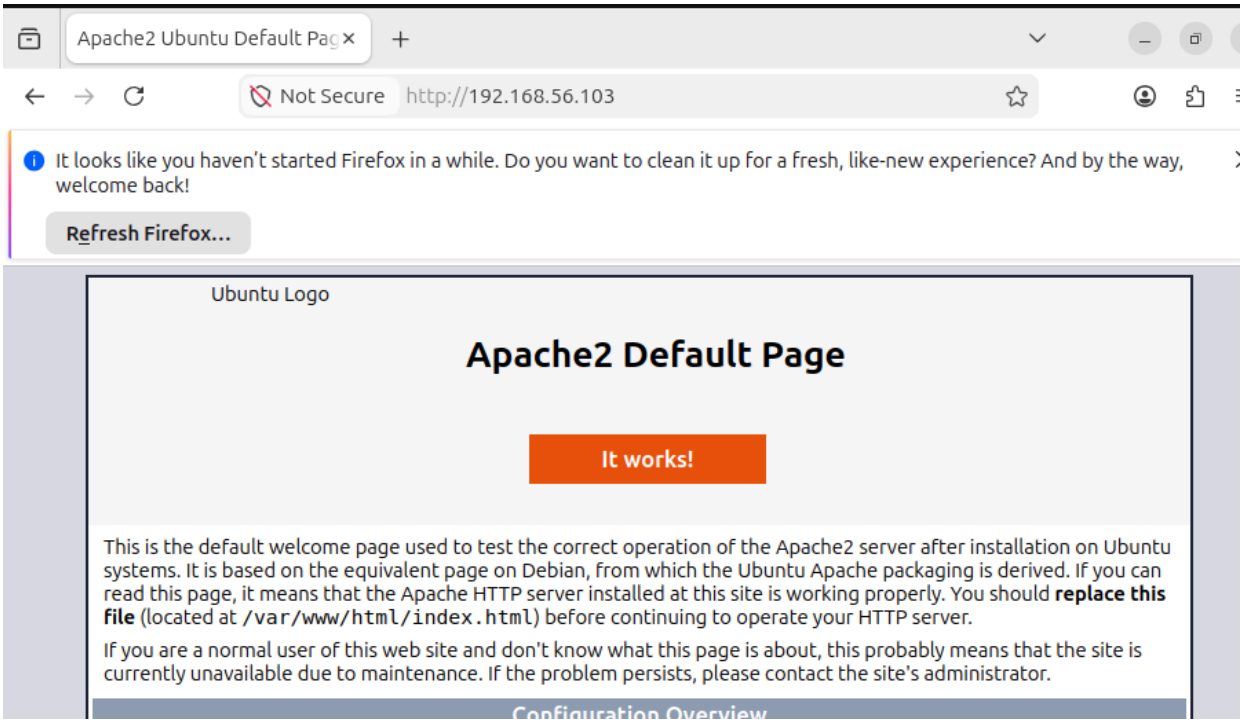
2.



- The task gathers facts if ubuntu 1 and 2 are connected to the inventory, while the next task installs apache for both the machines. Then it recaps stating that both machines are okay and have been changed (installation of apache2).

3.


Apache2 Ubuntu Default Pag
Not Secure
http://192.168.56.102
It looks like you haven't started Firefox in a while. Do you want to clean it up for a fresh, like-new experience? And by the way, welcome back!
Refresh Firefox...
Ubuntu Logo
Apache2 Default Page
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview


Apache2 Ubuntu Default Pag
Not Secure
http://192.168.56.103
It looks like you haven't started Firefox in a while. Do you want to clean it up for a fresh, like-new experience? And by the way, welcome back!
Refresh Firefox...
Ubuntu Logo
Apache2 Default Page
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

4.

```
masongsong@Masongsong:~/CPE212_Masongsong$ ansible-playbook --ask-become-pass install_a
pache.yaml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu1]
ok: [ubuntu2]

TASK [Install Apache Package] **************************************************************
[ERROR]: Task failed: Module failed: No package matching 'apachechebureche' is availabl
e
Origin: /home/masongsong/CPE212_Masongsong/install_apache.yaml:8:5

6
7
8   - name: Install Apache Package
      ^ column 5

fatal: [ubuntu1]: FAILED! => {"changed": false, "msg": "No package matching 'apachecheb
ureche' is available"}
fatal: [ubuntu2]: FAILED! => {"changed": false, "msg": "No package matching 'apachecheb
ureche' is available"}

PLAY RECAP *********************************************************************************
ubuntu1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=
0    rescued=0    ignored=0
ubuntu2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=
0    rescued=0    ignored=0

masongsong@Masongsong:~/CPE212_Masongsong$
```

- The output fails because it fails to recognize the package “apachechebureche” it is supposed to be a known package for it to install.

5.

```
masongsong@Masongsong: ~/CPE212_Masongsong
 GNU nano 7.2                                 install_apache.yaml
---
- hosts: all
  become: true
  tasks:


  - name: update repository index
    apt:
      update_cache: yes

  - name: install apache2 package
    apt:
      name: apache2
```

6.

```
masongsong@Masongsong:~/CPE212_Masongsong$ ansible-playbook --ask-become-pass install_a
pache.yaml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]
ok: [ubuntu2]

TASK [update repository index] *************************************************
changed: [ubuntu2]
changed: [ubuntu1]

TASK [install apache2 package] *************************************************
ok: [ubuntu2]
ok: [ubuntu1]

PLAY RECAP *********************************************************************
ubuntu1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0
ubuntu2                    : ok=3    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0

masongsong@Masongsong:~/CPE212_Masongsong$
```

- Yes, it outputs a changed for both machines in update repository index because it updated its cache.

7.

8.

```
masongsong@Masongsong:~/CPE212_Masongsong$ ansible-playbook --ask-become-pass install_a
pache.yaml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]
ok: [ubuntu2]

TASK [update repository index] *************************************************
changed: [ubuntu2]
changed: [ubuntu1]

TASK [install apache2 package] *************************************************
ok: [ubuntu1]
ok: [ubuntu2]

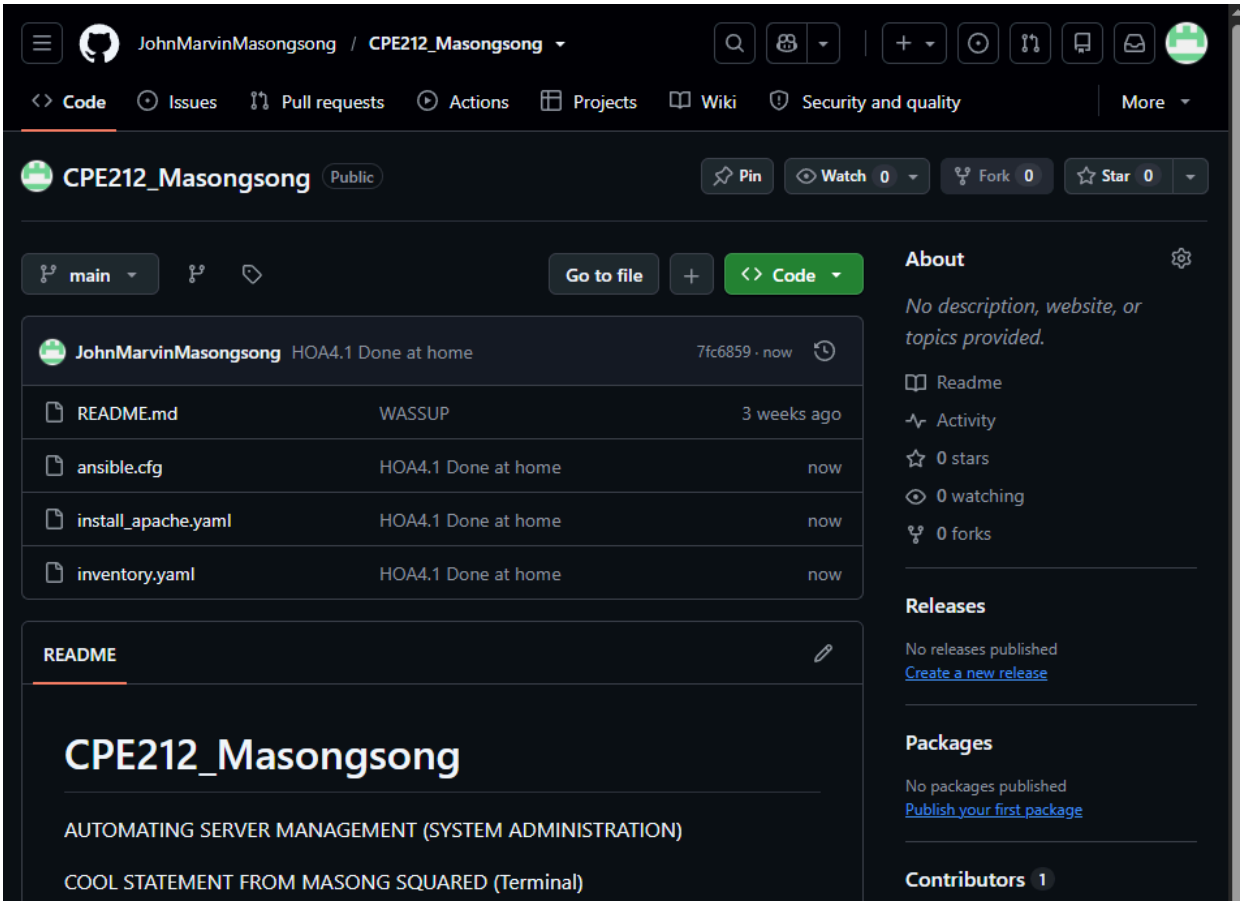
TASK [add PHP support for apache] **********************************************
changed: [ubuntu1]
changed: [ubuntu2]

PLAY RECAP *********************************************************************
ubuntu1                    : ok=4    changed=2    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=2    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0

masongsong@Masongsong:~/CPE212_Masongsong$
```

- The apache2-mod-php was installed by displaying changed on both machines.

9.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What is the importance of using a playbook?
   - Using a playbook allows you to bypass ad hoc commands which are just one liner commands that are usually done in the terminal, by using a playbook, you are able to put more than one line inside a text file, and run it with more than one commands linked to it.

2. Summarize what we have done on this activity.
   - We have done the basics of ad hoc commands to ensure that our playbooks and inventory.yaml files are working correctly. We also tried using playbooks to install different packages and verify them if they really are downloaded. We also talked about pushing all our work in our github which in return made us setup a new ssh key for github and my home pc.

Conclusion:
I have learned a lot from this HOA, but mainly the basics of ansible and how it works, we learned things like ad hoc commands which are one liner commands, and playbooks which utilizes the text file in order to execute multiple commands at once. We also had a refresh on how to connect our github with our ubuntu workstation from home. In addition, in the playbooks we were able to download and verify different packages that were installed. All in all, this HOA was really fun and I have learned a lot from it. I hope to use this in our future HOAs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

COMPUTER ENGINEERING

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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